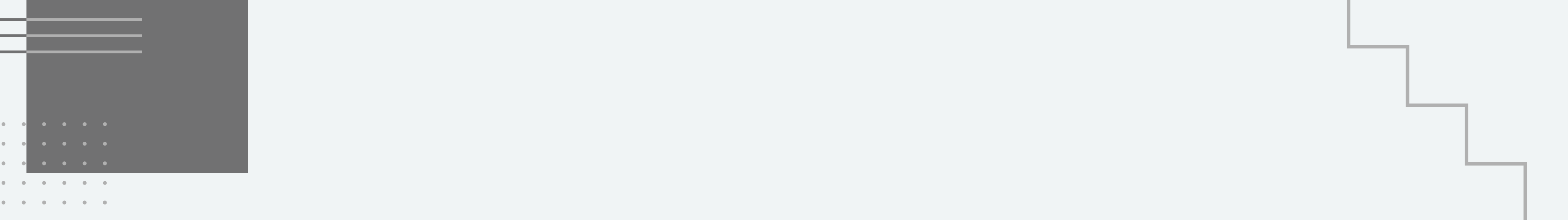




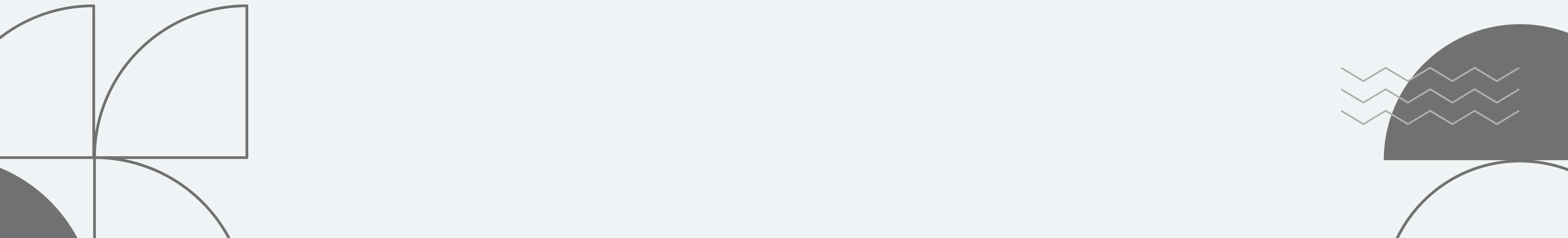
BOARDGAME ARE MADE OF RULES

Riccardo Conforto Galli



INTRODUCTION

The aim of this project is to evaluate how well different large language models process analyze and extract information from board-game rule texts.



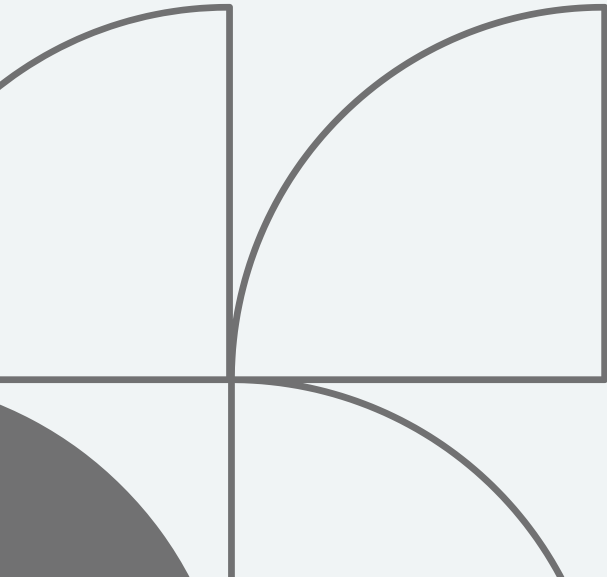
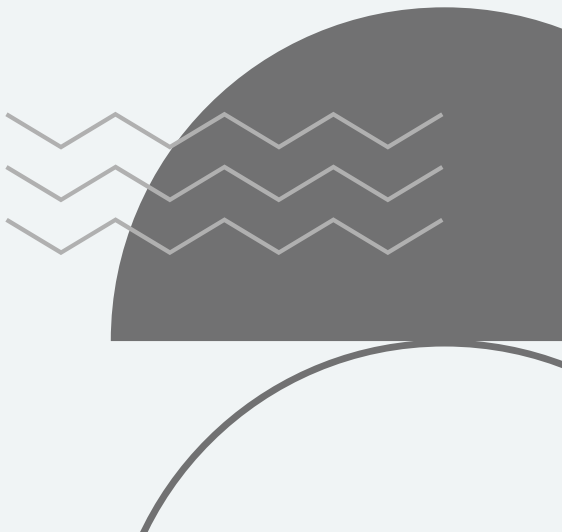
DATA

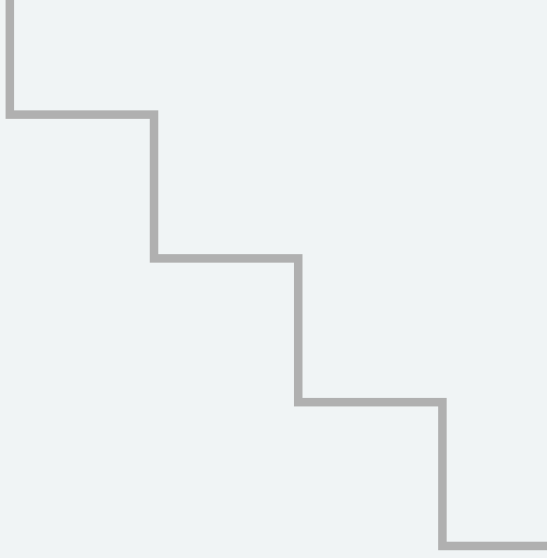
- 5 rule books in pdf format
- from BGG Hall of Fame
- complexity from light to medium

GAME	BGG-id
7 Wonders	316377
Catan	13
Dominion	209418
Power Grid	2651
Ticket to Ride	9209



RESEARCH QUESTION

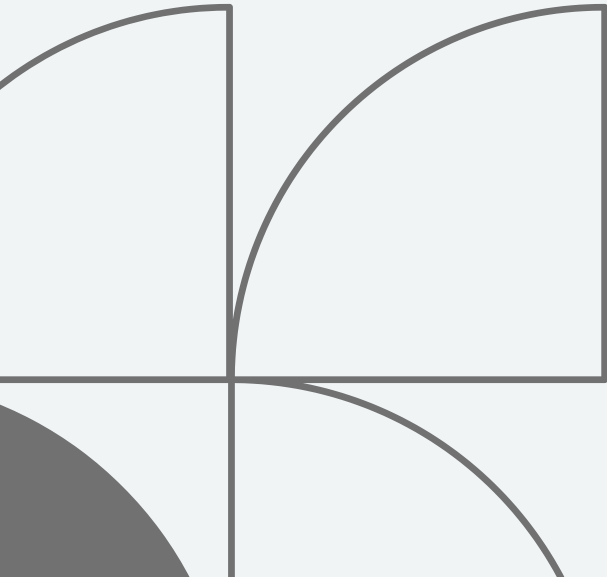
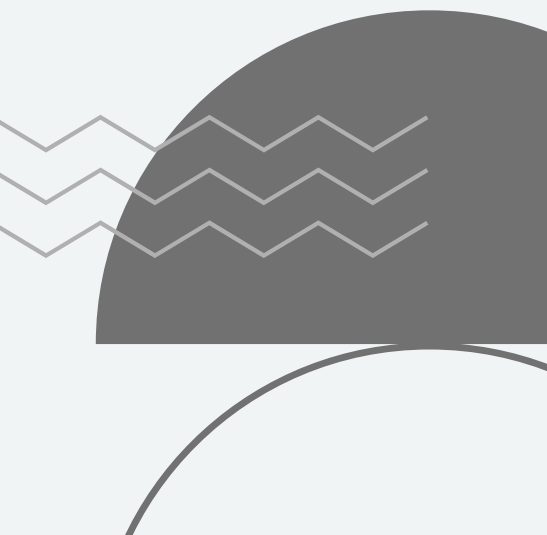
1. Ask the model to explain the game to different kind of audiences
 2. Feed the model a version of a game with missing or flawed rules
 3. Give the model a rule book and ask it to classify the mechanics and estimate the parameters
- 
- 

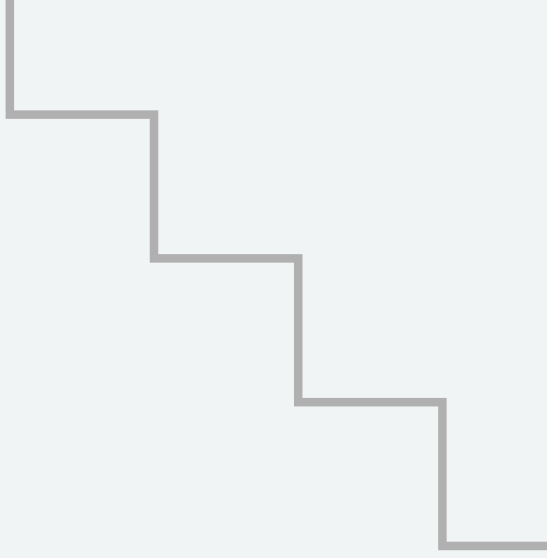


MODELS

run locally

temperature set to the recommended value if
present, 0.7 otherwise

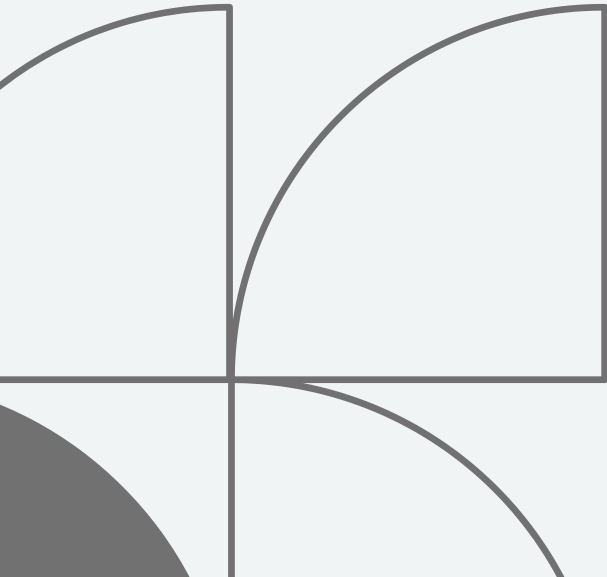
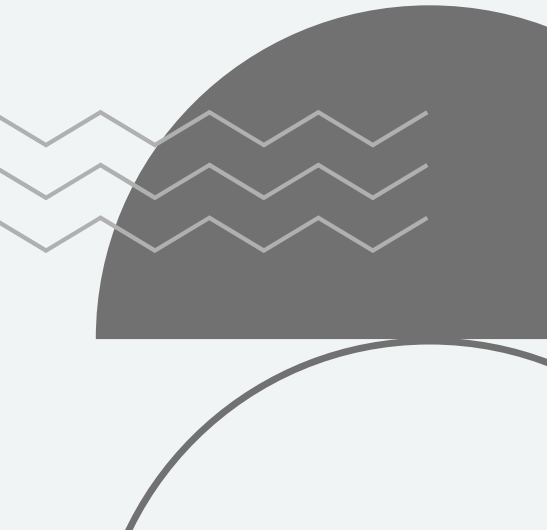
- 1.** LLaMA 3.1 8b
 - 2.** Qwen 2.5 7b
 - 3.** Gemma 3n 4b
- 
- 

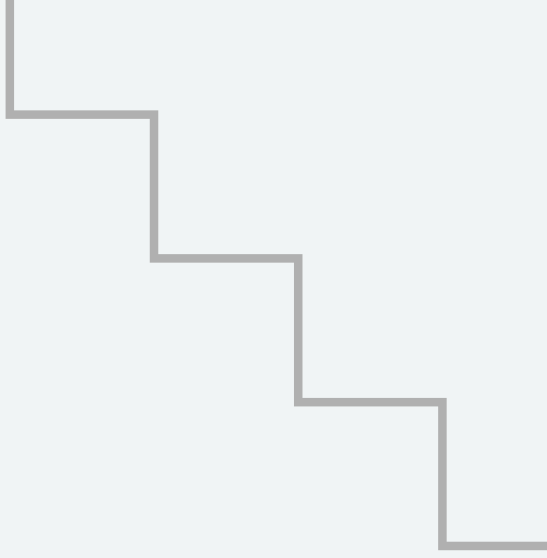


EXPLANATION

prompted to generate an explanation in conversation terms for subjects of age 7, 11 and 16 years.

Evaluated on:

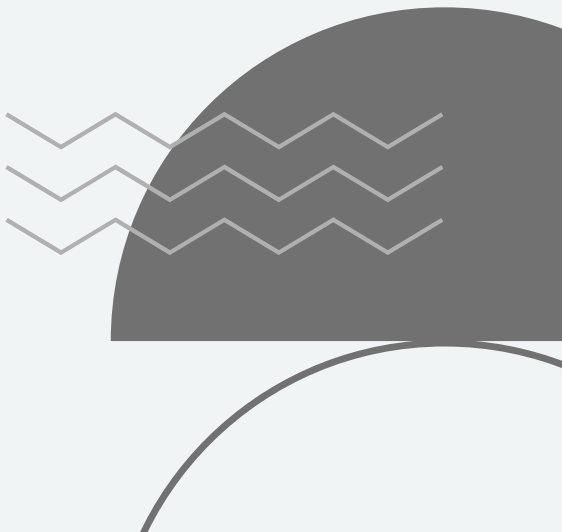
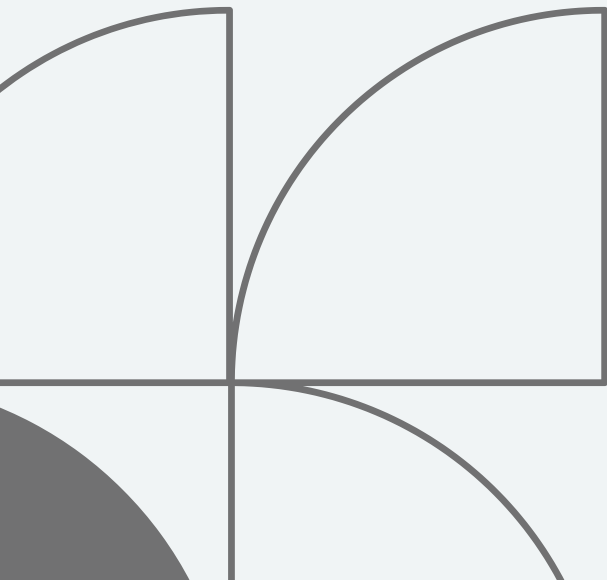
1. Readability
 2. Completeness
 3. Coinciseness
- 
- 



1. READABILITY

Three readability metric to evaluated generated outputs

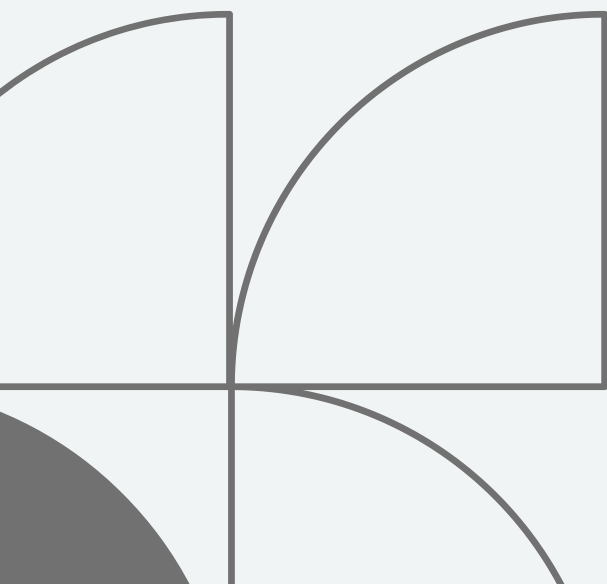
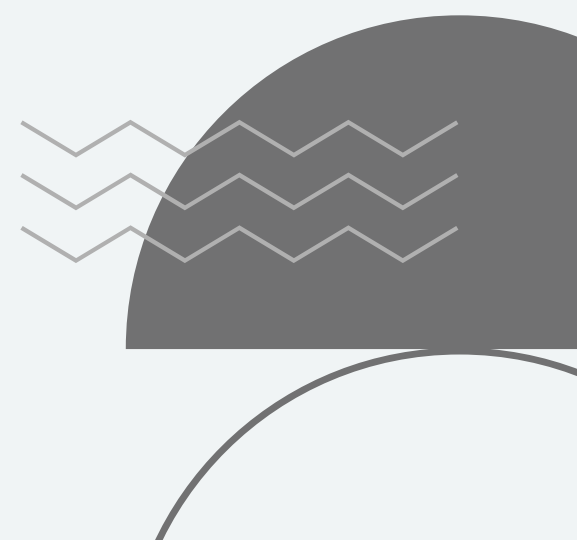
1. Flesch-Kincaid
2. SMOG
3. Dale-Chall



Model	Age	Expected	Flesch–Kincaid	SMOG	Dale-Chall grade	Dale–Chall
LLaMA	7	2	4.68	7.72	10-12	8.25
	11	6	5.76	8.47	10-12	8.46
	16	11	6.31	9.12	12	8.99
Qwen	7	2	4.41	7.09	10-12	8.13
	11	6	4.70	7.58	10-12	8.56
	16	11	5.35	8.25	>12	9.31
Gemma	7	2	3.68	6.97	9-10	7.70
	11	6	4.41	7.81	10	8.01
	16	11	4.81	8.08	10-12	8.36



2. COMPLETENESS

1. Extraction of rule hierarchy
 2. hierarchy restructuration
 3. detect rule presence
- 
- 

Completeness score

$$\frac{\sum_{r \in R} \mathbb{I}_E(r) \cdot v(r)}{\sum_{r \in R} v(r)}$$

$$v(r) = 4 - h(r)$$

detect rule presence

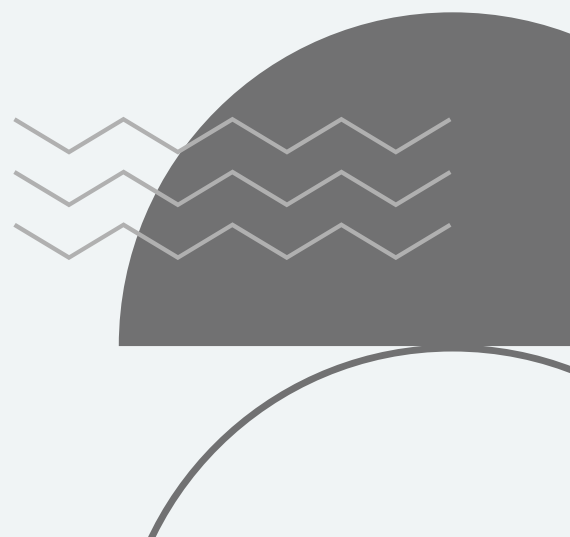
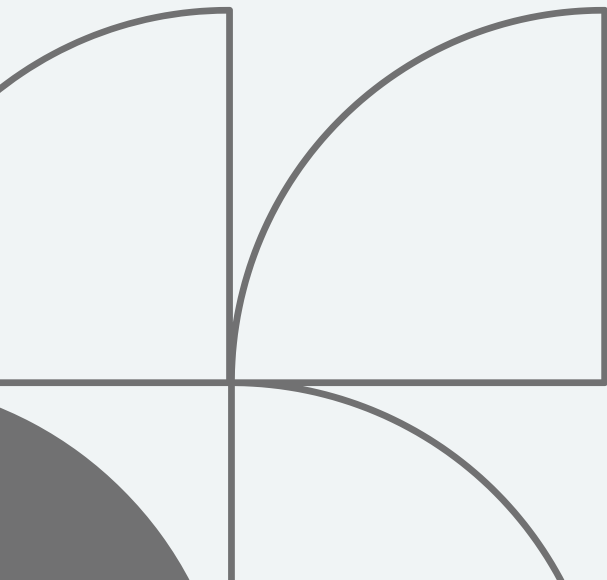
Sentence embedding applied to sliding windows.
Detected if cosine distance < 0.5

Hey there, young adventurer! Let's talk about the amazing game of Catan! ****What's the goal?**** The goal of the game is to collect 10 "victory points" (VPs). You earn VPs by building roads, settlements, and cities on the island of Catan. ****How do you win?**** You win the game when you have 10 or more VPs at the end of your turn. You can earn VPs by building settlements, cities, and roads, or by collecting special cards called "development cards". ****What happens in a turn?**** In a turn, you do two things: 1. ****Production phase****: You roll two dice to see which hexes on the island will produce resources (like brick, wood, or wheat). If you have a settlement or city on a hex that produces resources, you get to collect those resources! 2. ****Action phase****: You can use your resources to build roads, settlements, cities, or trade with other players. You can also play development cards to get special powers. ****Some cool things to remember:**** * When you roll a 7, you have to discard some of your resources and move the "robber" to a new hex. The robber will steal resources from another player who has a building on that hex! * You can trade resources with other players, but you can't give away cards you don't have. * When you build a road, settlement, or city, you have to follow some rules, like not building too close to other players' buildings. That's it! Catan is a super fun game that's all about building and trading. I hope you're excited to start playing!

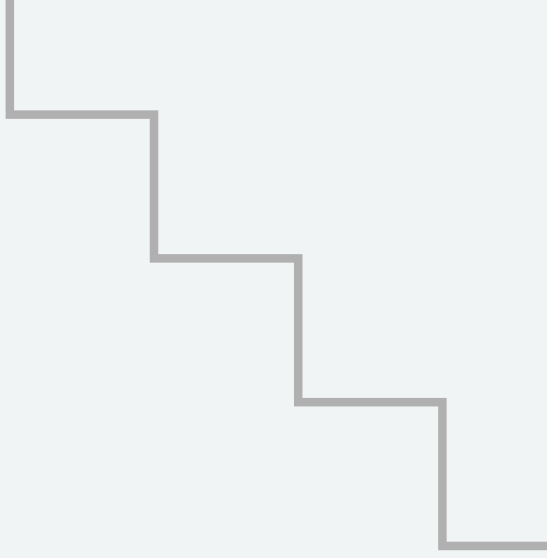
sliding window representation for: "Discard Resources on a Roll of 7"



3. COINCISENESS

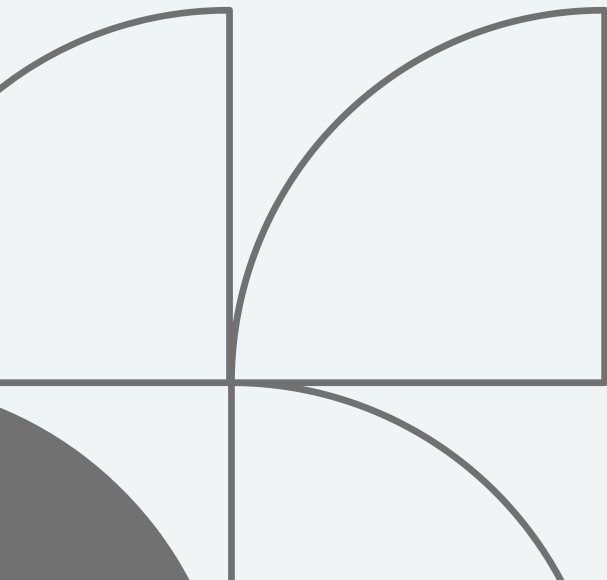
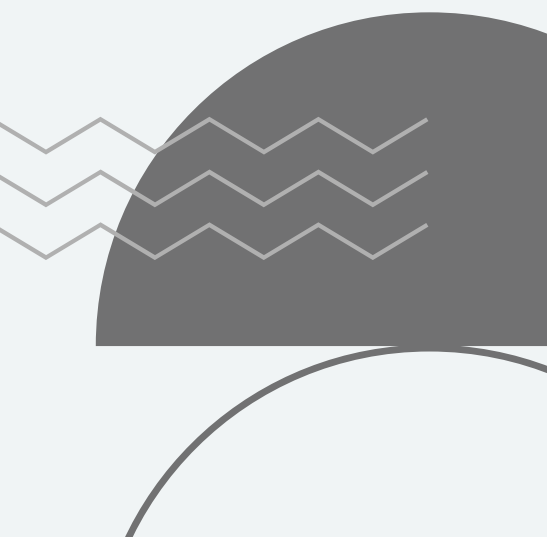
$$\frac{\textit{len}(\textit{explanation})}{\textit{len}(\textit{original})}$$


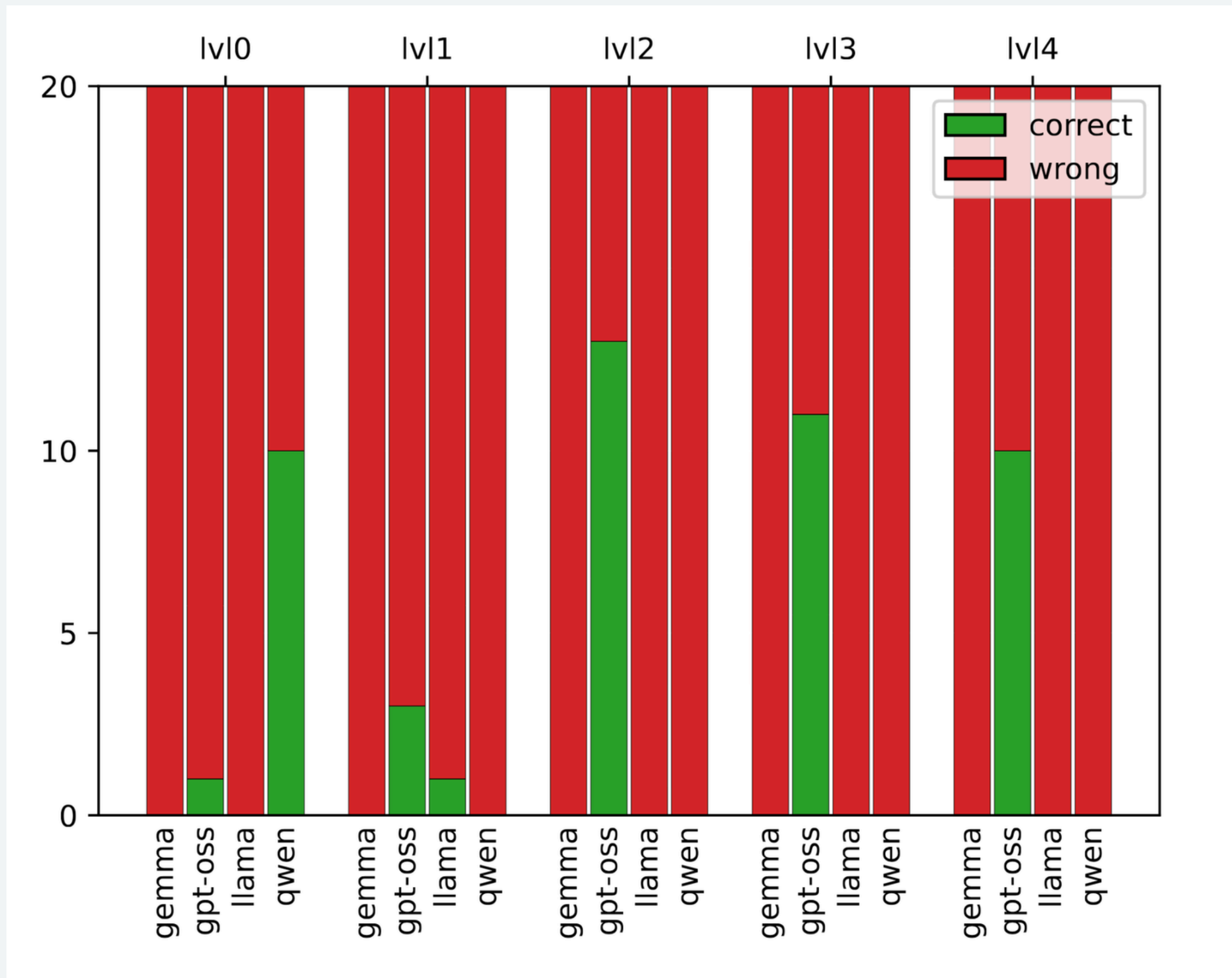
Game	Model	Rule coverage ↑	Relative Size ↓
7 Wonders	gemma	0.09	10.13
	llama	0.10	9.90
	qwen	0.07	10.68
Catan	gemma	0.17	11.86
	llama	0.18	12.28
	qwen	0.20	14.55
Dominion	gemma	0.39	6.61
	llama	0.36	6.75
	qwen	0.40	9.91
Power Grid	gemma	0.11	5.68
	llama	0.11	6.22
	qwen	0.10	5.50
Ticket to Ride	gemma	0.76	16.38
	llama	0.91	19.59
	qwen	0.90	23.32



ERROR DETECTION

five levels of edits:

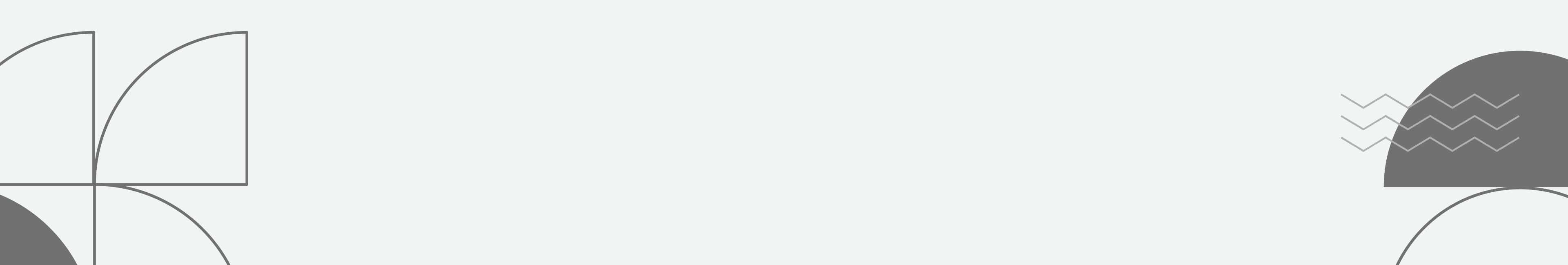
- 0.** Original
 - 1.** Missing
 - 2.** unsolvable
 - 3.** hard lock
 - 4.** unbalanced
- 
- 





PARAMETER ESTIMATION

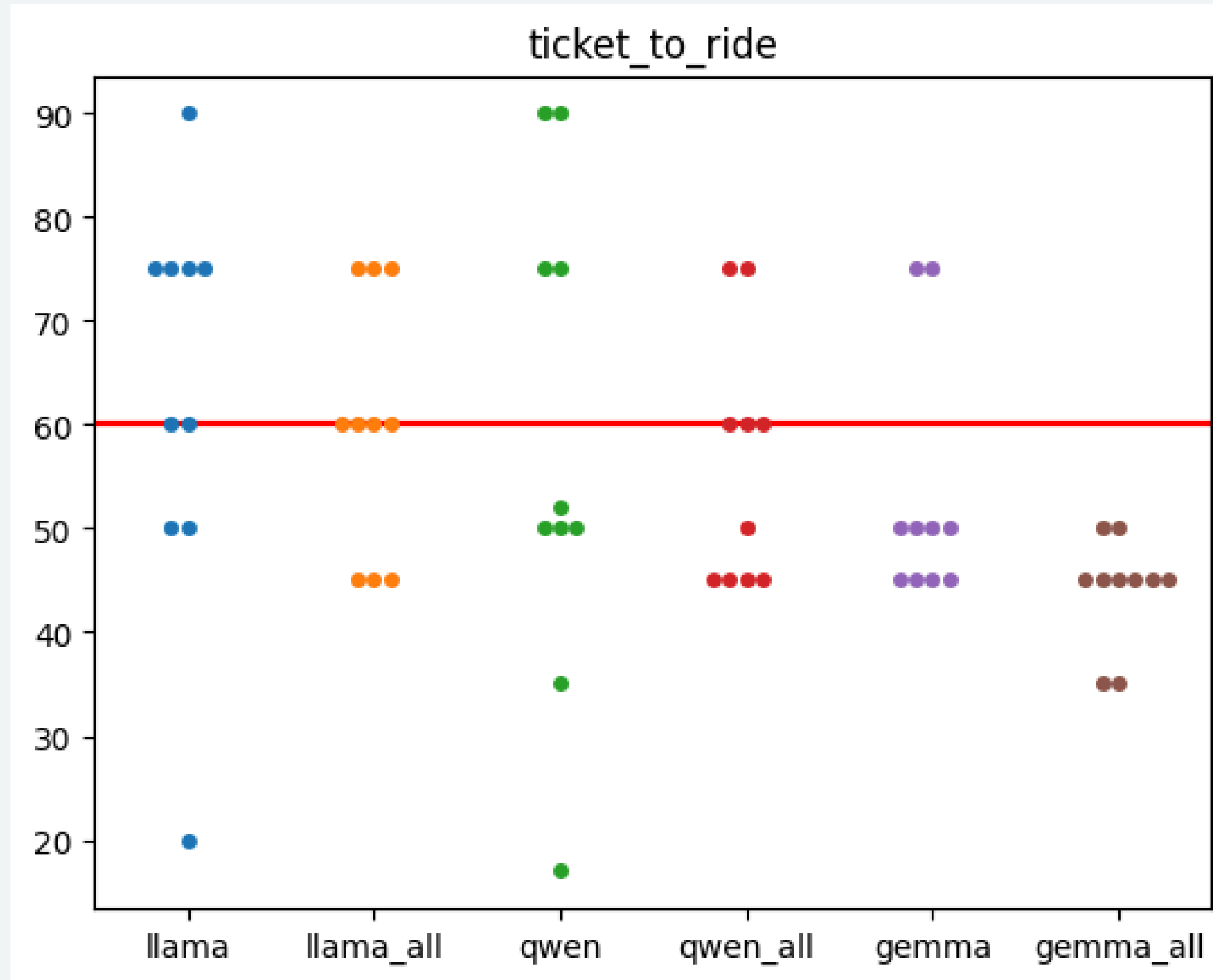
two different prompts:

- specific to each parameter
 - all parameters at once
- 

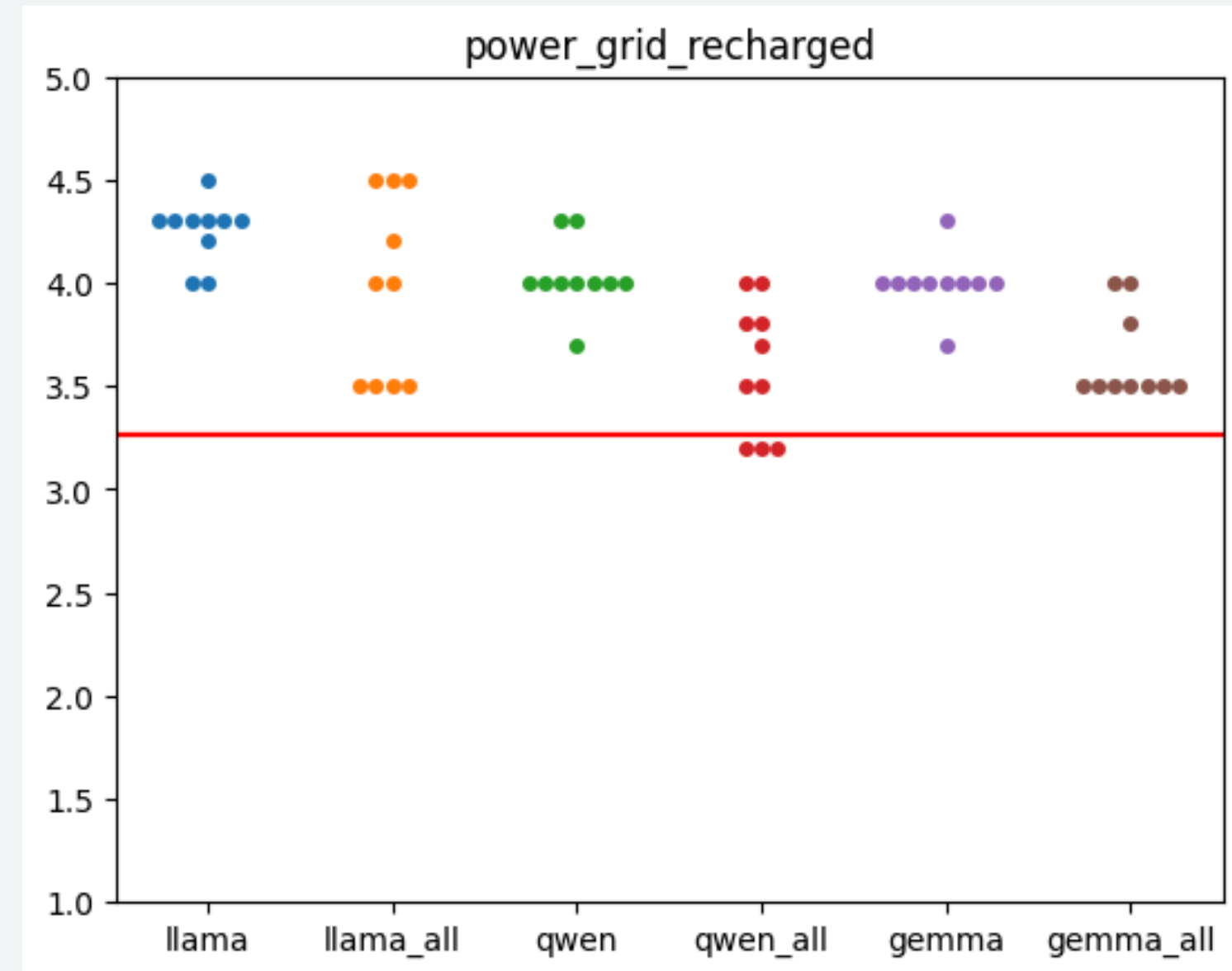
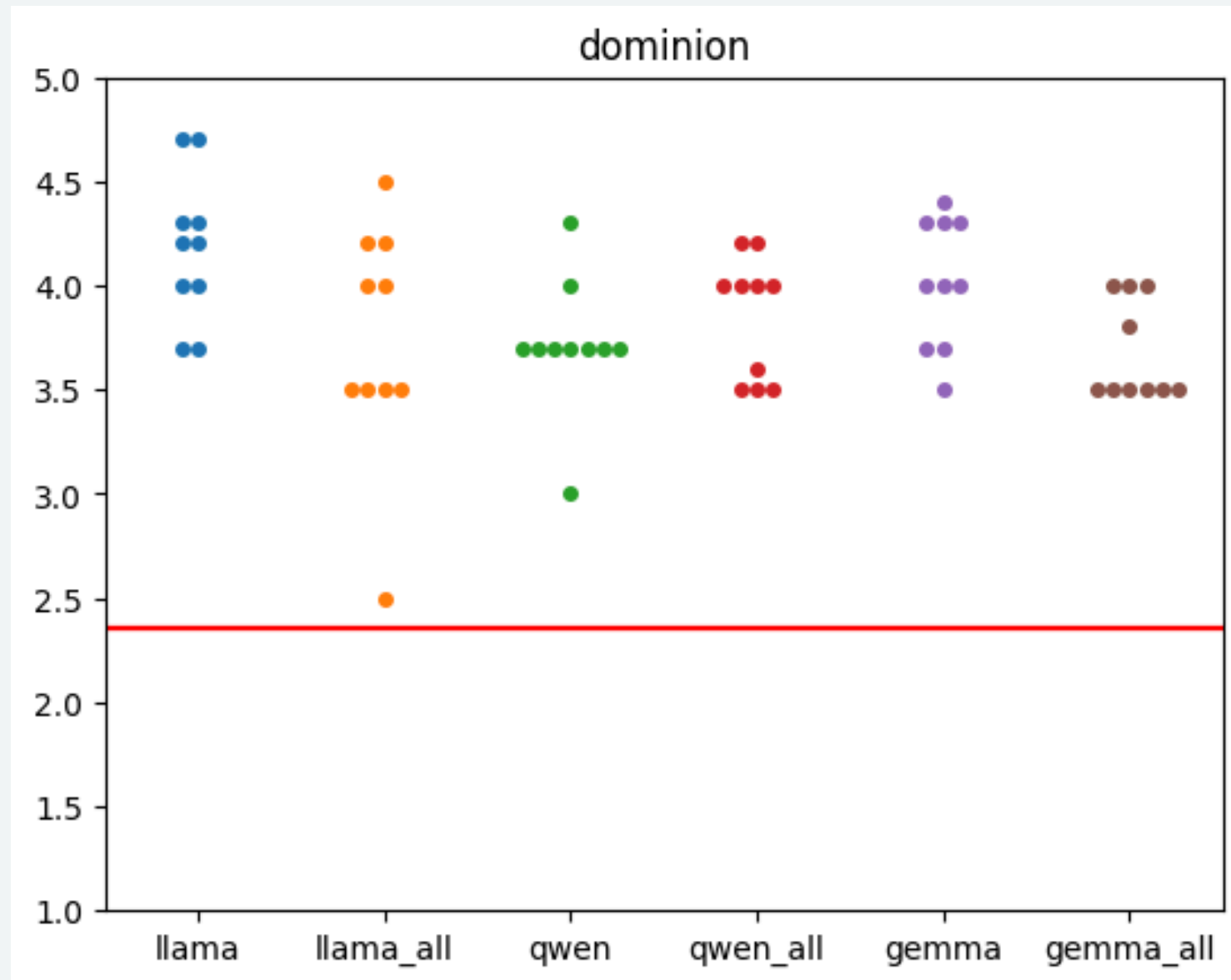
	Exact match		Loose match	
	mean precision	mean recall	mean precision	mean recall
gemma	0.144770	0.276071	0.256993	0.528810
llama	0.105373	0.119722	0.288261	0.351944
qwen	0.066857	0.059048	0.222447	0.230952

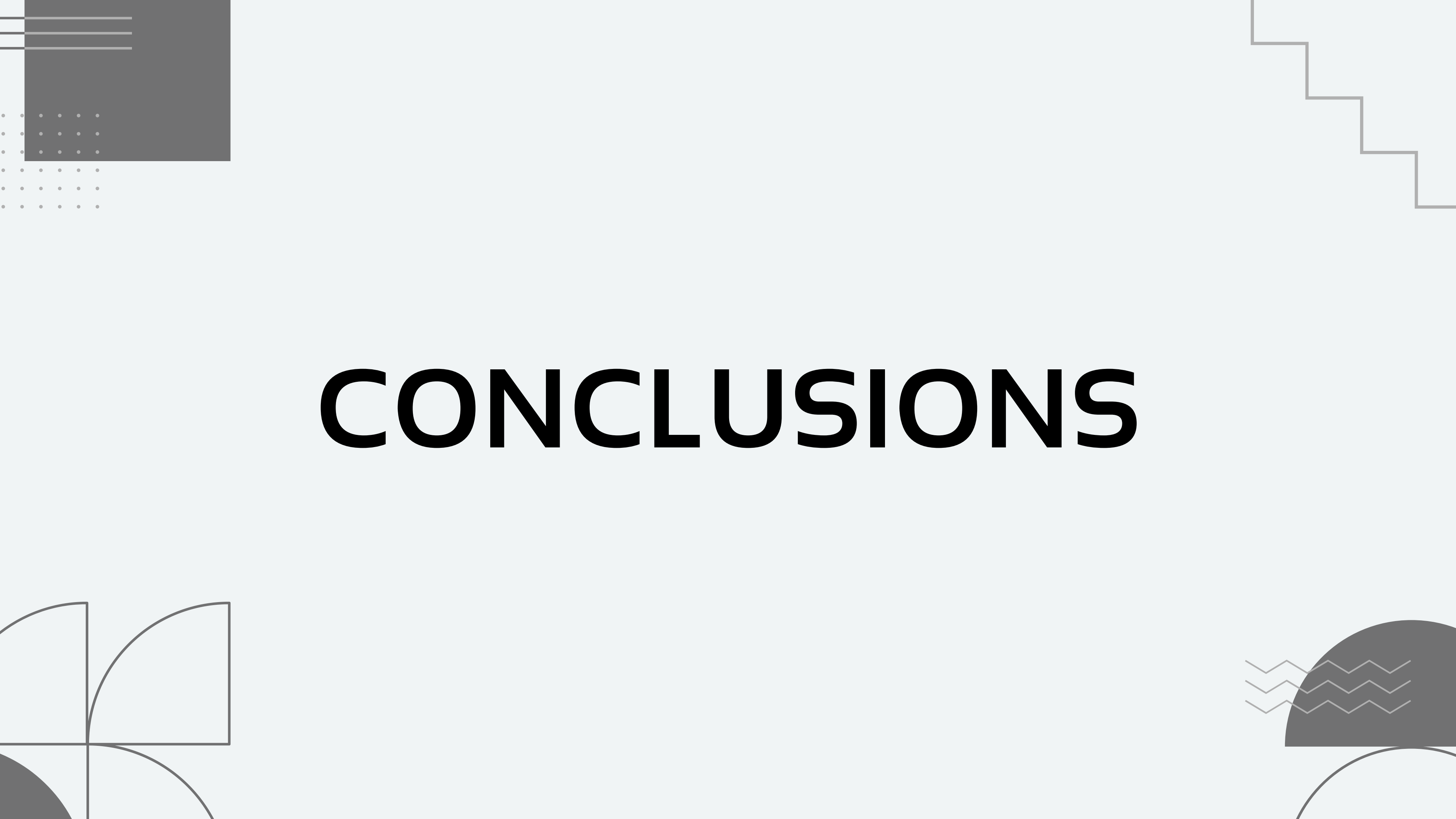
Model	Mean abs error		
	Complexity	Optimal players	Duration
llama	1.69	0.42	29.25
llama_all	1.23	1.05	33.00
qwen	1.22	0.30	32.40
qwen_all	1.06	0.58	24.38
gemma	1.09	0.33	20.38
gemma_all	1.05	0.48	35.88

Duration estimation is unreliable

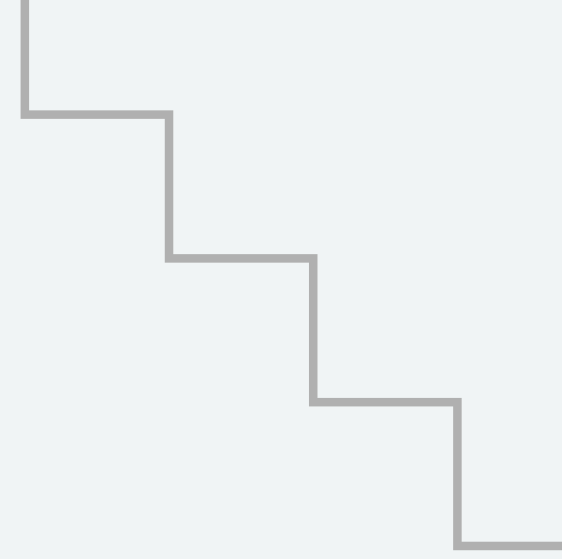


Complexity overestimation





CONCLUSIONS



**THANK
YOU**

